MITHIL KANSARA

Toronto, ON | **437-660-7815** | **mithilkansara8@gmail.com**
Mechanical Engineering Intern/Co-op

EDUCATION

Bachelor's in Engineering: ***Mechanical Engineering*** (Co-op Program Enrolled)
York University *- Toronto, ON | September 2023-Present (**4th Yr.**)*

SUMMARY

Fourth Year Mechanical Engineering student at York University with hands-on experience in CAD design, manufacturing, testing, and technical documentation through engineering internships, research, and multidisciplinary projects. Experienced in preparing drawings, BOMs, calculations, and technical reports while collaborating with teams in fast-paced manufacturing and lab environments. Strong analytical, organizational, and problem-solving skills developed through mechanical design, prototyping, and engineering-focused research initiatives.

PROFESSIONAL EXPERIENCE

STEM Program Mentor – *Co-op*

k2i Academy @ Lassonde School of Engineering | *Toronto, ON | May 2026 – Present*

- Researched AI applications in **manufacturing** & **quality assurance**, developing AI sensor-based solutions for *KPM Power*.
- Led **50**+ hands-on **STEM** workshops in **physics**, **electronics,** & **coding** while connecting **real engineering design practices**.
- **Collaborated** with educators & mentors to develop **Engineering**-aligned activities following *Ontario Education Standards*.
- Developed & **tested 20+** student project kits integrating AI cameras, sensors, *Micro*:*bit* coding-based automation workflows.
- **Independently managed** project documentation, detailed planning sheets, and student data regularly using ***Microsoft Excel***.

Center Assistant & Math-Reading Tutor – *Part-Time*

Kumon Canada Inc. | *Vaughan, ON | May 2025 – May 2026*

- Mentored **30**+ students daily, balancing tutoring, grading, and classroom support in a **fast-paced** structured **environment.**
- Monitored **classroom workflows**, maintaining accountability, organization, and adherence to Kumon procedures & standards.
- **Documented** student performance using ***Excel***, maintaining **accurate records** & **managing timelines** for daily preparation.
- **Communicated** student **progress** and learning gaps clearly to **supervisors** and parents through **written and verbal updates.**
- **Explained complex concepts clearly** while **adapting** quickly to **changing workloads** and varying student learning needs.

Engineering & CAD Design Intern – *Internship*

Pantheon Prototyping | *Toronto, ON | September 2024 – December 2025*

- Created **200**+ 3D printed parts and prototypes in **SolidWorks**, applying **GD&T** and **tolerance-based** design considerations.
- **Independently** operated **20 FDM & SLA 3D printers** in a **manufacturing environment**, with **consistent** print quality.
- Scheduled & **prioritized multiple project deadlines**, from managing CAD modeling to printing, packaging, and delivery.
- Documented **design metrics** and print data using **Excel** sheets & **Notion** for consistent workflow tracking & organizing.
- Communicated with **100**+ **customers** and team members to **troubleshoot designs & improve manufacturing performance**.

RELEVANT SKILLS

- SolidWorks
- 3D Modeling & Design
- Microsoft Office
- AutoCAD
- CREO
- Inventor/Fusion360
- LabVIEW/MATLAB
- Engineering Drawings
- BOMs & Technical Documentation
- GD&T & Tolerance Analysis
- Machining & Manufacturing
- 3D Printing & Prototyping
- FEA & CFD Analysis
- Safety and Quality Standards
- Written & Verbal Communication
- Adaptability
- Leadership
- Multitasking
- Organization
- Team Collaboration
- Problem-Solving

PROJECTS

Formula 1 Rear Wing Design & Simulation

York University | September 2025 – December 2025

Designed, analyzed, and validated a Formula 1 rear wing under strict geometry, deflection, and structural compliance requirements.

- Performed **CFD and FEA** in Siemens StarCCM+ under multiple load cases, conducting mesh refinement and convergence studies to validate aerodynamic loads and confirm structural compliance before finalizing results.
- **Designed within strict geometry and deflection limits** in SolidWorks, iterating on mesh density and load cases to balance downforce and structural performance within required boundaries.
- Documented solver settings, boundary conditions, and validation steps in a **structured technical report**, earning above a 90% project grade.

3D Printed RC Car & Arduino Integration

York University | January 2025 – April 2025

A remote-controlled car designed to transport objects, built and programmed from scratch using 3D printed components and an Arduino.

- **Designed and 3D printed a chassis**, steering mechanism, and multi-stage gearbox in SolidWorks, tuning gear ratios to balance torque and speed for carrying loads.
- Wired an **Arduino** Nano to a motor driver, receiver module, and servo steering, programming control logic to handle throttle, direction, and turning via remote input.
- Developed **technical documentation and lab reports**, using Excel to record and analyze performance data across different gear configurations.

3D Printed Robotic Arm and LabVIEW Integration

York University | September 2024 – December 2024

A multi-axis robotic arm designed to lift and sort objects using sensor feedback and programmed motion control.

- Designed and 3D printed a multi-axis robotic arm in SolidWorks, refining joint geometry and wall thickness through **iterative prototyping** to handle repeated lifting loads.
- **Programmed motion sequences in LabVIEW** with an Arduino UNO controlling servos at each joint, using sensor feedback to detect, pick up, and sort blocks by position.
- Performed **load testing** at each joint, performing **FEA** by identifying weak points under stress, and reinforcing critical sections to improve reliability during repeated operation.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Mechanical Design Team Member | York University Robotics Society

Lassonde School of Engineering | *September 2023 – Present*

- Assembled mechanical components in SolidWorks for a Mars Rover, supporting integration & manufacturable designs.
- Designed & **machined** parts in the machine shop, following technical drawings, tolerances, & safety procedures.
- Collaborated with **multidisciplinary team members** to prototype, fabricate, troubleshoot, and refine mechanical systems.

Steering & Braking Team Member | YorkU GNCTR

Lassonde School of Engineering | *September 2024*

- Designed an Ackermann steering system & ratchet-pawl braking mechanism for a race toboggan in **SolidWorks**, with detailed 2D **AutoCAD drawings** and **Bill of Materials documentation**.
- **Performed calculations and FEA** to verify stress and deflection, ensuring designs met competition **safety standards.**
- Managed design through **fabrication and testing phases**, contributing to a **4th and 3rd place national finish**

Competition Mentor & Judge | York Engineering Competition

Lassonde School of Engineering | *September 2024*

- Mentored student teams during engineering challenges, guiding their approach and keeping them on track.
- Supported participants by answering questions, clarifying problems, and **encouraging teamwork** and problem solving.
- Evaluated submissions & **delivered constructive feedback**, helping students identify strengths & areas for improvement.